

EXERCICE

Décomposer en éléments simples dans $\mathbb{R}(X)$ la fraction rationnelle suivante :

$$F = \frac{X}{(X-1)^3(X-2)}.$$

$$\deg(F) = 1 - 4 = -3 < 0$$

$$F = \frac{a_1}{X-1} + \frac{a_2}{(X-1)^2} + \frac{a_3}{(X-1)^3} + \frac{b}{X-2}$$

$$\text{avec } b = \left[\frac{X}{(X-1)^3} \right]_{X=2} = \frac{2}{(2-1)^3} = \boxed{2}$$

$$a_3 = \left[\frac{X}{X-2} \right]_{X=1} = \frac{1}{1-2} = \boxed{-1}$$

$$F = \frac{X}{(X-1)^3(X-2)} = \frac{a_1}{X-1} + \frac{a_2^{??}}{(X-1)^2} + \frac{-1}{(X-1)^3} + \frac{2}{X-2}$$

1] on multiplie F par $(X-1)^3$:

$$\frac{X}{X-2} = a_1(X-1)^2 + a_2(X-1) - 1 + \frac{2(X-1)^3}{X-2}$$

2] on dérive: $\left(\frac{X}{X-2}\right)' = \frac{(X-2-X)}{(X-2)^2} = 2a_1(X-1) + a_2 + 6(X-1)^2 \dots (*)$

3] on remplace X par 1: $\left(\lim_{X \rightarrow 1}\right)$

$$\frac{-2}{(1-2)^2} = a_2 \Rightarrow \boxed{a_2 = -2}$$

on dérive (*):

$$\left[\frac{-2}{(X-2)^2}\right]' = \frac{-2(X-2)(-2)}{(X-2)^4} = \frac{4}{(X-2)^3} = 2a_1 + 12(X-1) \dots$$

on remplace X par 1: $\frac{4}{(1-2)^3} = 2a_1 \Rightarrow \frac{4}{-1} = 2a_1 \Rightarrow \boxed{a_1 = -2}$

Donc :

$$F = \frac{-2}{x-1} + \frac{-2}{(x-1)^2} + \frac{-1}{(x-1)^3} + \frac{2}{x-2}$$

$$F = \frac{x+5}{(x-1)^4(x+2)^3}$$

$$= \frac{a_1}{x-1} + \frac{a_2}{(x-1)^2} + \frac{a_3}{(x-1)^3} + \frac{a_4}{(x-1)^4} + \frac{b_1}{x+2} + \frac{b_2}{(x+2)^2} + \frac{b_3}{(x+2)^3}$$

Diagram illustrating the partial fraction decomposition of the rational function $F = \frac{x+5}{(x-1)^4(x+2)^3}$.

The decomposition is shown as a sum of terms:

- $\frac{a_1}{x-1}$
- $\frac{a_2}{(x-1)^2}$
- $\frac{a_3}{(x-1)^3}$
- $\frac{a_4}{(x-1)^4}$
- $\frac{b_1}{x+2}$
- $\frac{b_2}{(x+2)^2}$
- $\frac{b_3}{(x+2)^3}$

Annotations:

- Red arrows and circled numbers (1, 2, 3, 4) indicate the order of terms for the "méthode du crochet" (bracket method).
- The term $\frac{a_4}{(x-1)^4}$ is highlighted in yellow.
- The terms $\frac{b_1}{x+2}$, $\frac{b_2}{(x+2)^2}$, and $\frac{b_3}{(x+2)^3}$ are highlighted in green.
- The term $\frac{b_3}{(x+2)^3}$ is also highlighted in yellow.
- The text "méthode du crochet" is written in red above the decomposition.
- The text "dérivée" (derivative) is written in red above the term $\frac{a_4}{(x-1)^4}$.